CSE606: Computer Graphics / Term 1 (2026-27) / Programming Assignment 1 (A1)
International Institute of Information Technology Bangalore

Announcement Date: Sep 30, 2026 (Wed)
Submission Deadline: 11:59 pm IST, Nov 01, 2026 (Sun)

Summary: Rendering and manipulation of 3D models. (15% of final grade)
Learning Objectives:

- Creating 3D models (using a modeling tool)
- Importing 3D mesh models
- Transformations of 3D objects
- Computing animation paths and transforms
- View transformations
- Picking model objects

Group Assignment: Labyrinth Builder

The goal of this assignment is for a team to develop an interactive labyrinth design and demonstrate how the player navigates through the maze.

3D Models: The maze is built out of modular wall and floor blocks arranged on a 3D grid. The maze should have a start point and an end point.

The maze must have 3-5 obstacles that are objects rendered using surface meshes from third-party sources (e.g., Stanford Bunny, Armadillo, etc.) or generated in Blender[1]. The goal of the player is to find a path to the end without passing through the obstacles, i.e., the player should turn back and retrace steps upon approaching an obstacle. Hence, the goal of the designer is to ensure that such a path exists.

Viewing Modes: The player can switch between three views:

1. Top View/Design Mode: to design the maze in 2D by clicking and rearranging walls.
2. World View/Design Mode: to render the maze and objects (player and obstacles) in 3D.
3. Player View/Play Mode: to navigate through the 3D maze.

In the top view, the player and obstacles can be placed using colored circles. In the player view, the player need not be rendered, as the view is ego-centric for the player, like the fly-through in a scene.

Transformations: Incorporate 3D transformations, i.e., scaling, rotation, and translation for both the player (i.e., camera) and obstacles. Use “object picking” in design modes to apply transformations to the objects.

[1] Extra marks for using models generated on Blender with sufficient complexity.

Transformations can be designed based on the implementation of the movements within the maze. A few suggestions:

- Translation of the player would mean moving forward or backward, and the same for the obstacles (some obstacles can be considered to be “live”).
- Rotation of the player has to be intuitive, e.g., a z-axis rotation to turn back; y-axis rotation to look up and down; and x-axis rotation to look side to side. The rotation of the obstacles could mean any in-place/instance rotations, depending on the logic of the mobility of the concerned obstacle.
- Scaling could be exclusively done for the obstacle as an instance transformation within available space.

Additional Effects: There can be special effects to indicate the start and end of the player’s quest in the maze. This can be a simple indicator, such as a change in the color of the floor tile ahead of the player in player view. Implementation of lighting is not required for this assignment

Requirements: The 3D transformations must be added as matrices. Rotation must be implemented using quaternions and a virtual trackball. The movement of the player has to be controlled by deciding the translation amount for every iteration. Interactions using the mouse are needed for all 3D transformations, and using the keyboard, for changing viewing modes. There should be an option to show/hide principal axes (x,y,z axes) at the reference point.

The key-action binding for user interactions must be intuitive. For example, the up and down arrows can increase/decrease the speed of the player, left and right arrows can scale the object up or down, “X” and “x” can increase/decrease the rotation about the x-axis, etc. The key bindings should be kept unique, so that these operations can be done in any order and may be interleaved.

Deliverables: Submissions must be made on LMS.

1. The deliverables in a zipped folder must include a source code folder, a folder with screenshots, a demo video not longer than 5 minutes, and a brief report describing key aspects of the assignment and answers to the questions below.
2. If the deliverables are larger than the maximum allowable size on LMS, submit a text document with a repository URL (Google Drive, OneDrive, etc.). Care must be taken to not change this repository until the end of the semester. Access to this repository to anyone with the link must be activated. Managing access is the onus of the student team.
3. Questions for the report:
    a. To what extent did you reuse code from A1?
    b. How did you enable 3D graphics on WebGL?
    c. What is the rationale for the concatenation of transformations in your implementation?
    d. How are obstacles added or deleted in the maze? How are walls rearranged?

Grading Rubric:
(Out of 30 points, to be scaled to 15 points in final grade)

- 10 marks for all functionalities, with a focus on completeness of the assignment
- 8 marks for code
- 4 marks for report
- 8 marks for video

Resources:
PLY file archives for meshes:

- https://people.sc.fsu.edu/~jburkardt/data/ply/ply.html
- https://sites.cc.gatech.edu/projects/large_models/
- https://graphics.stanford.edu/data/3Dscanrep/